r/aiwars

Energy consumption + environmental impact from AI generation?

Posted by u/btcprox • • 0 points • 20 comments

Recently saw some online posts floating about arguing that, at least in the realm of images, AI-generated pictures may have larger negative environmental impact (e.g. in energy consumption) compared to completely manually creating the digital image with the relevant software

I admit I haven't really delved much into researching this, so I'm not entirely sure about actual measured energy consumption in the processes of either training or using the AI models, but I do know that training large models necessarily demands high computing power

But I'm also uncertain if human artists necessarily get a pass either, for one since certain media are also intuitively resource intensive, e.g. 3D modeling and printing

If we constrain the scope to just the delivery of one quality image, it could be argued that the human would spend more time labouring over the process, so perhaps that equates to a larger footprint

Yet usually those who showcase quality AI-generated images, very likely would have needed to spend additional time curating and iterating by adjusting parameters, so that could also lead to a larger footprint

Any additional thoughts on either side on this?

Comments

u/chillaxinbball • 1 points • 2024-11-30 06:04 UTC

Using the exact same hardware takes way more power to manually create an image than have and ai model generate the image.

Let's be conservative and say it takes me a few hours to make an amazing looking image in a combination of tools like Octane, unreal, and Photoshop. Many of these tools take a lot of resources to run. You can run the GPU rather heavily during that time period.

In contrast, you can run the same GPU for 11 seconds using an AI model and usully get a better looking render.

So what's the impact? Less than what we have now.

Granted this is a one off situation. Realistically the artist would be working the same amount of time to refine their image how they see fit. The Ai is more of a tool to improve that process.

If energy consumption is the main concern, perhaps be more worried about games using the exact same card for hours on end and not producing anything of worth.

u/themfluencer • 1 points • 2024-11-29 18:23 UTC

Data centers and collection consumes vast amounts of energy. Cooling these computers takes a lot of water. Localized processes are better for the environment than globalized processes. For example: fruits. Eating a fruit grown in your city is always going to be more environmentally friendly than eating a fruit grown halfway across the world.

u/EngineerBig1851 • 2 points • 2024-11-29 18:07 UTC

Ai generator runs on the same hardware you play fortnite on.

Even then diffusion models work in bursts - image is created, spike in energy consumption as your videocard gets to work, and rhen back down when your videocard Isn't needed. In videogames graphics card is needed 100% of the time.

Arguing AI somehow "drinks water" while majority run it locally is fucking stupid.

u/Tyler_Zoro • 2 points • 2024-11-29 16:23 UTC

> Recently saw some online posts floating about arguing that, at least in the realm of images, AI-generated pictures may have larger negative environmental impact (e.g. in energy consumption) compared to completely manually creating the digital image with the relevant software

Every such claim I've seen is either backed up by one informal paper from Huggingface, which was horrifically flawed (they asserted that each image generated consumes enough power to charge a cell phone fully... which would mean that my home rig is melting itself down every time it produces an image in 10 seconds), or it's based on assessments of training costs without amortizing those costs across every generation.

Feel free to cite something new in this realm. I'd love to read it.

u/Primary_Spinach7333 • 2 points • 2024-11-29 15:32 UTC

This isn’t an issue solely unique to ai, just computers in general - all that energy could have the potential to come from a clean energy source and overall this is an absurd claim.

Also why did you post this if you said you had not done research on it? Sorry, just saying.

u/WelderBubbly5131 • 2 points • 2024-11-29 15:31 UTC

I have used this copy-pasted comment many times. It might get tiring or irritating, but all articles mentioned here are peer reviewed by scientists, professors and those who have excelled in their fields.

AI is significantly less pollutive compared to humans: https://www.nature.com/articles/s41598-024-

54271-x

Published in Nature, which is peer reviewed and highly prestigious: https://en.m.wikipedia.org/wiki/Nature_%28journal

>AI systems emit between 130 and 1500 times less CO_2e per page of text compared to human writers, while AI illustration systems emit between 310 and 2900 times less CO_2e per image than humans.

Data centers that host AI are cooled with a closed loop. The water doesn't even touch computer parts, it just carries the heat away, which is radiated elsewhere. It does not evaporate or get polluted in the loop. Water is not wasted or lost in this process.

"The most common type of water-based cooling in data centers is the chilled water system. In this system, water is initially cooled in a central chiller, and then it circulates through cooling coils. These coils absorb heat from the air inside the data center. The system then expels the absorbed heat into the outside environment via a cooling tower. In the cooling tower, the now-heated water interacts with the outside air, allowing heat to escape before the water cycles back into the system for re-cooling."

Source: https://dgtlinfra.com/data-center-water-usage/

Data centers do not use a lot of water. Microsoft's data center in Goodyear uses 56 million gallons of water a year. The city produces 4.9 BILLION gallons per year just from surface water and, with future expansion, has the ability to produce 5.84 billion gallons (source: https://www.goodyearaz.gov/government/departments/water-services/water-conservation). It produces more from groundwater, but the source doesn't say how much. Additionally, the city actively recharges the aquifer by sending treated effluent to a Soil Aquifer Treatment facility. This provides needed recharged water to the aquifer and stores water underground for future needs. Also, the Goodyear facility doesn't just host AI. We have no idea how much of the compute is used for AI. It's probably less than half.

gpt-4 used 21 billion petaflops of compute during training (https://ourworldindata.org/grapher/artificial-intelligence-training-computation) and the world uses 1.1 zetaflop per second (https://market.us/report/computing-power-market/ per second as flops is flop per second). So from these numbers (21 * 10^9 * 10^15) / (1.1 * 10^21 * 60 * 60 * 24 * 365) gpt-4 used 0.06% of the world's compute per year. So this would also only be 0.06% of the water and energy used for compute worldwide. That's the equivalent of 5.3 hours of time for all computations on the planet, being dedicated to training an LLM that hundreds of millions of people use every month.

Using it after it finished training costs HALF as much as it took to train it: https://assets.jpmprivatebank.com/content/dam/jpm-pb-aem/global/en/documents/eotm/a-severe-case-of-covidia-prognosis-for-an-ai-driven-us-equity-market.pdf

(Page 10)

Image generators only use about 2.9 W of electricity per image, or 0.2 grams of CO_2 per image: https://arxiv.org/pdf/2311.16863

For reference, a good gaming computer can use over 862 Watts per hour with a headroom of 688 Watts: https://www.pcgamer.com/how-much-power-does-my-pc-use/

One AI image generated creates the same amount of carbon emissions as about 7.7 tweets (at 0.026

grams of CO_2 each, totaling 0.2 grams for both). There are 316 billion tweets each year and 486 million active users, an average of 650 tweets per account each year: https://envirotecmagazine.com/2022/12/08/tracking-the-ecological-cost-of-a-tweet/

https://www.nature.com/articles/d41586-024-00478-x

“ChatGPT, the chatbot created by OpenAI in San Francisco, California, is already consuming the energy of 33,000 homes” for 13.6 BILLION annual visits plus API usage (source: https://www.visualcapitalist.com/ranked-the-most-popular-ai-tools/). that's 442,000 visits per household, not even including API usage.

From this estimate (https://discuss.huggingface.co/t/understanding-flops-per-token-estimates-from-openais-scaling-laws/23133), the amount of FLOPS a model uses per token should be around twice the number of parameters. Given that LLAMA 3.1 405b spits out 28 tokens per second (https://artificialanalysis.ai/models/gpt-4), you get 22.7 teraFLOPS (2 * 405 billion parameters * 28 tokens per second), while a gaming rig's RTX 4090 would give you 83 teraFLOPS.

Everything consumes power and resources, including superfluous things like video games and social media. Why is AI not allowed to when other, less useful things can?

In 2022, Twitter created 8,200 tons in CO_2e emissions, the equivalent of 4,685 flights between Paris and New York. https://envirotecmagazine.com/2022/12/08/tracking-the-ecological-cost-of-a-tweet/

Meanwhile, GPT-3 (which has 175 billion parameters, almost 22x the size of significantly better models like LLAMA 3.1 8b) only took about 8 cars worth of emissions (502 tons of CO_2e) to train from start to finish: https://truthout.org/articles/report-on-chatgpt-models-emissions-offers-rare-glimpse-of-ais-climate-impacts/

By the way, using it after it finished training costs HALF as much as it took to train it: https://assets.jpmprivatebank.com/content/dam/jpm-pb-aem/global/en/documents/eotm/a-severe-case-of-covidia-prognosis-for-an-ai-driven-us-equity-market.pdf

(Page 10)

> **u/themfluencer** • 1 points • 2024-11-29 18:25 UTC
>
> Do you know if the water used to cool computers is potable or not? I would be interested to know. If they’re using grey water I’d be fine with it. But superfluous water for machines when safe water is already a scarce resource concerns me.
>
> > **u/WelderBubbly5131** • 2 points • 2024-11-29 18:46 UTC
> >
> > According to the 3rd link, only large companies like Google can afford to use portable water for their data centers. Other, smaller ones (almost all of them), use (and reuse) grey/reclaimed water or just air cooling.
> >
> > Google is also trying tho, 25% of their water cooled systems are reclaimed/grey water based.
> >
> > Almost ALL of them (some exceptions) use closed circuits for their water cooling, so the consumption is renewed at a less alarming rate too.
> >
> > > **u/themfluencer** • 1 points • 2024-11-29 18:49 UTC

I can rock with grey water for cooling. I am a nerd for sustainable long term growth of technology- It often serves us better than unfettered unlimited growth. But the problem is that our markets are run by folks who cannot think beyond “number go up=economy good”. Thus the negative externalities of their growth gets pushed on to future generations and the bottom billion.

u/No-Opportunity5353 • 3 points • 2024-11-29 15:15 UTC

Generating an image uses up like 1/100 of the electricity a human will use to draw it on a computer and tablet over hours or days.

> **u/ryenaut** • 0 points • 2025-04-18 23:31 UTC
>
> This is a moot point when people are generating images at an exponentially higher volume, speed, and frequency than someone drawing it manually.

u/duckrollin • 3 points • 2024-11-29 14:38 UTC

Like any massed computing it uses a lot of electricity, but that electricity can be from renewable sources, or at least from an efficient power plant with carbon capture. Same reason we're trying to swap to electric cars around the world right now.

However, it's about 0.01% of CO2 emissions, compared to 16% from cars and 24% from transport in total. ICE car engines are also never going to be efficient compared to a power station.

So in conclusion: Yes, it's bad for the environment, so is your hairdryer and charging your phone. If you want to reduce CO2 emissions then dump your ICE car and get a bike.

u/m3thlol • 11 points • 2024-11-29 14:04 UTC

One for one an AI system uses less energy per output than a human does, *after it's trained*. If we factor in training, then yes training obviously uses a lot of energy, however once a model is trained it's usable forever (again at a lower emission rate than a human).

Basically if you factor the training cost into cost per output then it gets lower with each output.

https://www.nature.com/articles/s41598-024-54271-x

> **u/Phemto_B** • 1 points • 2024-11-29 19:43 UTC
>
> Yep. The "problem" that AI has is that it centralizes most of the energy so that we can SEE it, instead of just distributing it out until it's invisible. If we commission a 100 million drawings, just ignore the energy used by millions of cintiqs all over the world, running for 2 billion user-hours (at about 70watts --> 140 Mwh). But if a single data center burns a megawatt hour to train a model that ultimately gets used for 100 million images, we can watch the dial on the meter spinning and have a nice panic about it.

> **u/eziliop** • 4 points • 2024-11-29 15:04 UTC
>
> > *after it's trained*
>
> This is an important distinction to make, and one antis like to gloss over. I swear it felt like talking to a brick wall when I tried explaining this.
>
> Anti: "AI is bad for environment because it uses so much power!"

Me: "Well during training it certainly does, but after that it's literally peanuts even compared to power usage we daily use on mundane, everyday stuffs. Not to mention training is a one-time thing and in theory we'll have a model that can last until the end of time."
Anti: "But, but it uses so much power so it's bad for the environment!"

Yeah... 🤷‍♂️

u/Josseph-Jokstar • 1 points • 2024-11-30 03:27 UTC

Okay, but someone who uses AI can and usually will generate hundreds of images compared to a digital artist, wouldn't that change the power usage ratio?

u/Pretend_Jacket1629 • 1 points • 2024-11-30 06:24 UTC

not for the equivalent task.

plus, you have to understand how incredibly low the energy is to generate an image. we're talking about less than 20 minutes of your computer being idle, or 3 second of any other high intensive software running. we're talking about 2.9 W. your toaster in one single push to make a slice of toast in the morning uses 172 to 620 times as much electricity as a single image takes.

u/Josseph-Jokstar • 1 points • 2024-11-30 12:20 UTC

Idk I can still see ur comment.

Thanks for clarifying.

Anyhow what about this e-waste thing people been talking about? Is that also another exaggerated issue?

u/sleepy_vixen • 1 points • 2024-11-30 12:37 UTC

E-waste is about physical electronics and their disposal. It's a contentious issue and there's a lot to be said about the unnecessary production volume and subsequent landfill junk of certain electronics but nothing really to do with AI in particular.

u/eziliop • 1 points • 2024-11-30 10:47 UTC

I can see your comment just fine, and thanks for addressing the question with hard data.

u/Phemto_B • 1 points • 2024-11-29 19:35 UTC

Yep. If you make a time plot the CO2 emissions that have resulted from form solar panel production, it's an exponential curve that shows no sign of slowing down. Clearly solar panel production is a massive threat to the environment that will kill us all!